C EXPERIMENTS

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C.1 Experimental Setup

Performing concept erasure using `PEF` does not require any parameter estimation except when the distributions are unequal. In this scenario, we use the Bayesian Optimization library (Nogueira, 2014–) and use the upper confidence bound (UCB) of exploitation and exploration as the acquisition function (with $\kappa = 2.5$). For evaluation of $\mathcal{V}$-information and MSE loss, we use the MLPClassifier and MLPRegressor functions from the scikit-learn (Pedregosa et al., 2011) library with its default settings. All the baseline models were trained on a 22GB NVIDIA Quadro RTX 6000 GPU and experiments were performed using the PyTorch (Paszke et al., 2019) framework. For baseline methods, we use the hyperparameters presented in the corresponding papers.

Next, we discuss the data generation process for the synthetic experiments reported in Section 5. We consider two concept groups and sample N 3D representations from a uniform distribution for each group. These representations form the support of the probability distributions we define next. We either define an equal (uniform or Gaussian) or unequal probability distribution using the support elements from each group. Using these distributions, we sample 10K representations from each group to form our final representation set. The concept label for each representation is the group from which it was sampled. For toxicity classification, we use the Jigsaw dataset, which contains online comments and the label is a fraction between 0 and 1 indicating the toxicity of the comment. We consider online comments associated with the following religion groups – Buddhist, Christian, Hindu, Jewish, Muslim, and others.

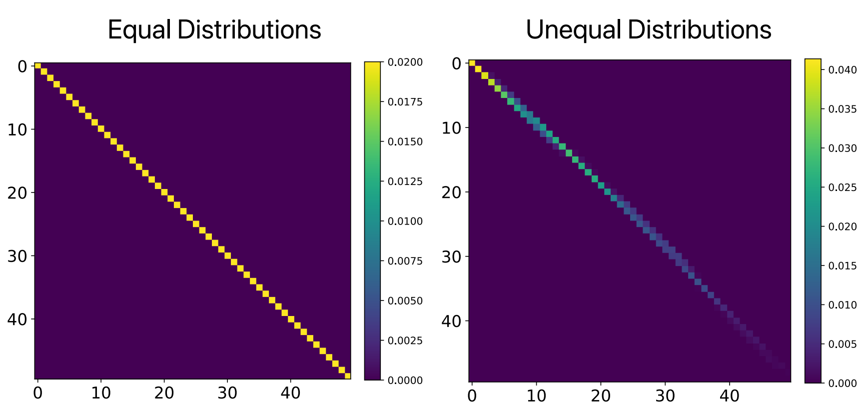

Figure 7: (*Left*) We show the minimum entropy coupling (joint distribution) between two equal distributions. We observe that the coupling map is a diagonal matrix indicating a 1-1 bijective map. (*Right*) We show that minimum entropy coupling between two unequal distributions is not a bijective map and individual support elements can be mapped to different elements.

C.2 Analysis Experiments

We conduct analysis experiments to inspect the erasure functions obtained using the formulation in Eq. 4. Specifically, we visualize the coupling maps (joint distributions) formed between two distributions $\Gamma(P_i, Q)$. We consider two scenarios: (i) equal distributions ($P_i = Q$) and (ii) unequal distributions ($P_i \neq Q$). In Figure 7, we visualize the minimum entropy coupling maps formed in these two settings. In Figure 7 (left), we observe that the coupling map is bijective (1-1 map), as predicted by Lemma 4.2. In Figure 7 (center), we observe that the coupling map is not bijective and therefore perfect erasure is not achieved.

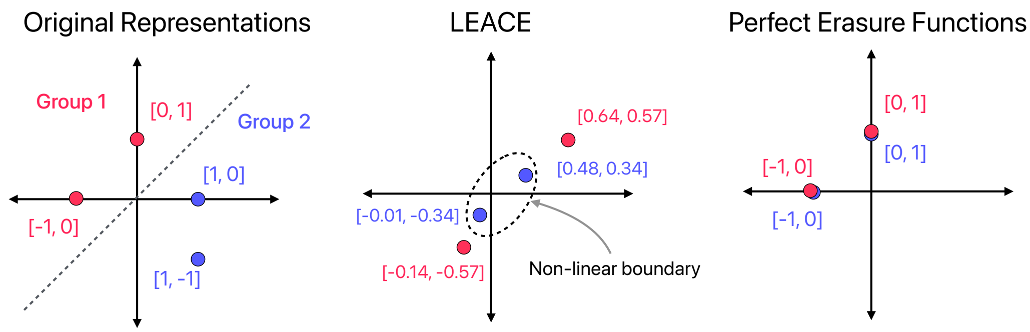

Figure 8: A toy example to illustrate the efficacy of erasure techniques LEACE and PEF. (*Left*) The original representation space with two linearly separable concept groups. (*Middle*) The erased representations obtained using LEACE, where there still exists a non-linear boundary separating the two groups. (*Right*) The erased representations from PEF where the concept has been perfectly erased.

C.3 Comparison with LEACE

In this section, we compare our proposed method PEF with LEACE (Belrose et al., 2024). LEACE is another perfect erasure technique that prevents linear adversaries from extracting concept information from erased representations. However, stronger non-linear adversaries may still be able to extract concept information. We illustrate this using a toy example in Figure 8. In this example, we consider a 2D original representation space (left plot in Figure 8) with two concept groups (each with only two elements). After erasure using LEACE (Figure 8 (middle)), we observe that there still exists a non-linear boundary that can separate the two groups and thereby reveal concept information. In contrast, PEF precisely maps elements between the two groups (Figure 8 (right)), making it impossible for any adversary to distinguish them. This example illustrates a scenario where LEACE can reveal concept information while PEF remains effective.

D BROADER IMPACT

Erasing sensitive concepts from data representations can reduce bias and enhance privacy, but it may also lead to the loss of valuable information, potentially diminishing the effectiveness of a machine learning model. The definition of sensitive concept attributes varies widely across cultural, ethical, and legal contexts. This work assumes that these attributes can be clearly defined and universally agreed upon, which is not always feasible. Therefore, developers should consider the societal impact carefully before implementing such erasure frameworks in practice.

`PEF` is intended to be used in applications where the developer is aware of the concept that needs to be erased. `PEF` can only erase concepts where the labels are annotated either as categorical attributes. One potential misuse of `PEF` would be to define relevant features for a task (e.g., educational background) as a concept to be erased. In such scenarios, the decision-making system can end up relying on arbitrary personal information to make hiring decisions. It is important to have oversight over such practices by using standard fairness measures like demographic parity.